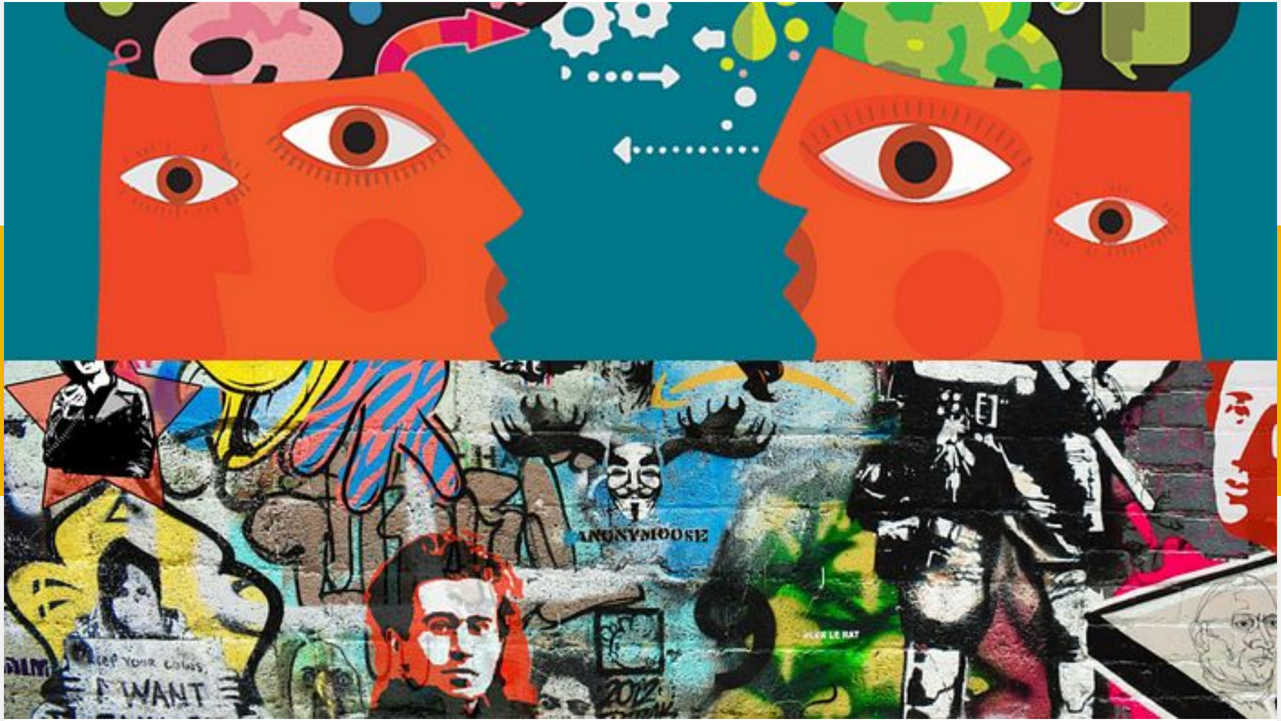

Demystifying Language with AI

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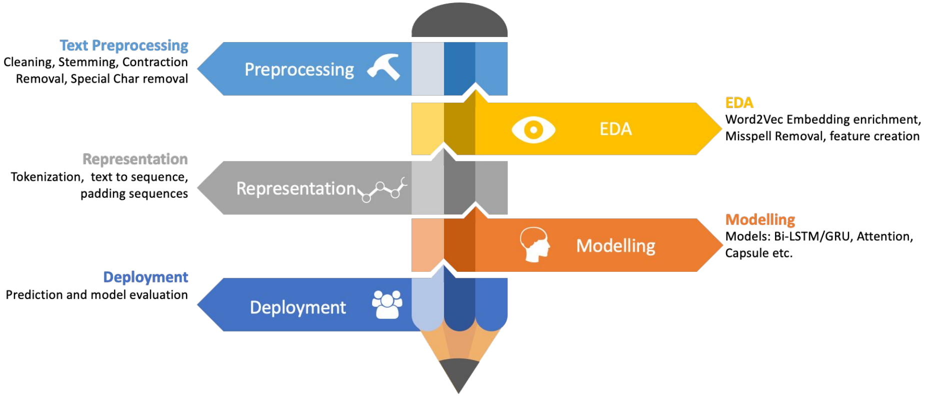
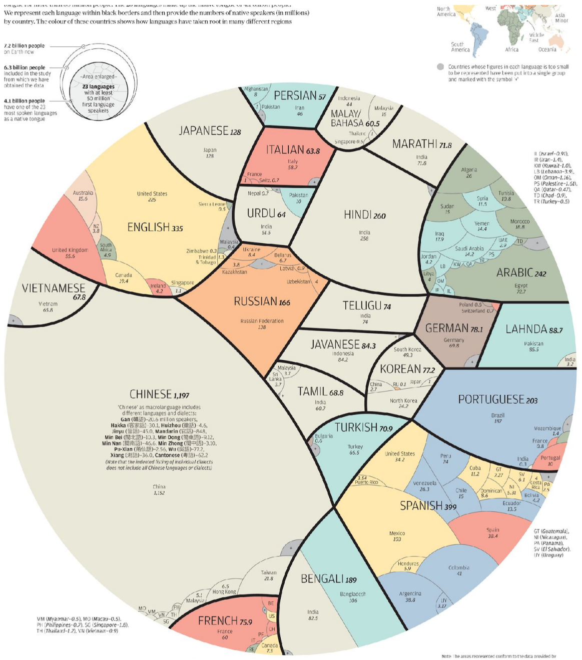
Objectives

- Field Overview
- Background
- Building blocks of text
- Preparing Text for Automation



What is Natural Language Processing?

- Language is complex, varied, ambiguous, and **unstructured**.
  - Over 7000 known languages
- Study of automating languages
  - Intersection of Computer Science and Linguistics
  - Tough job of finding algorithmic commonalities among the cognitive diversity of the many known languages of the world.
- Two concomitant line of work
  - Natural Language Understanding (NLU)
  - Natural Language Generation (NLG)



Natural Language Processing (NLP) is a field of study that is primarily focused upon developing algorithms that gives computers the ability to interpret, manipulate, and comprehend human language.

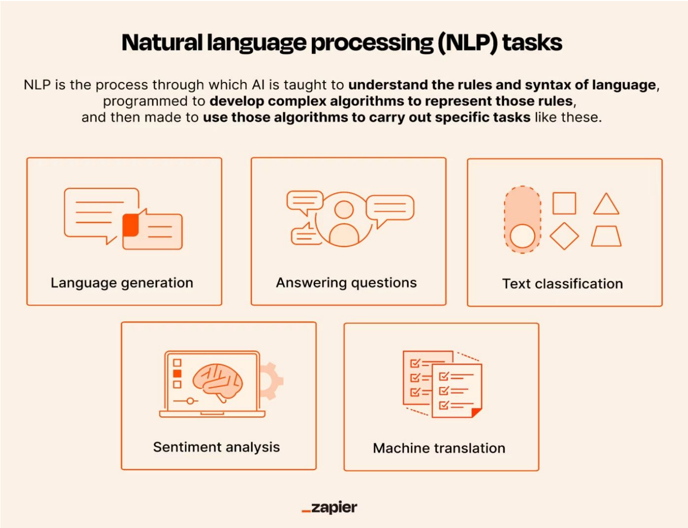
What is Natural Language Processing?

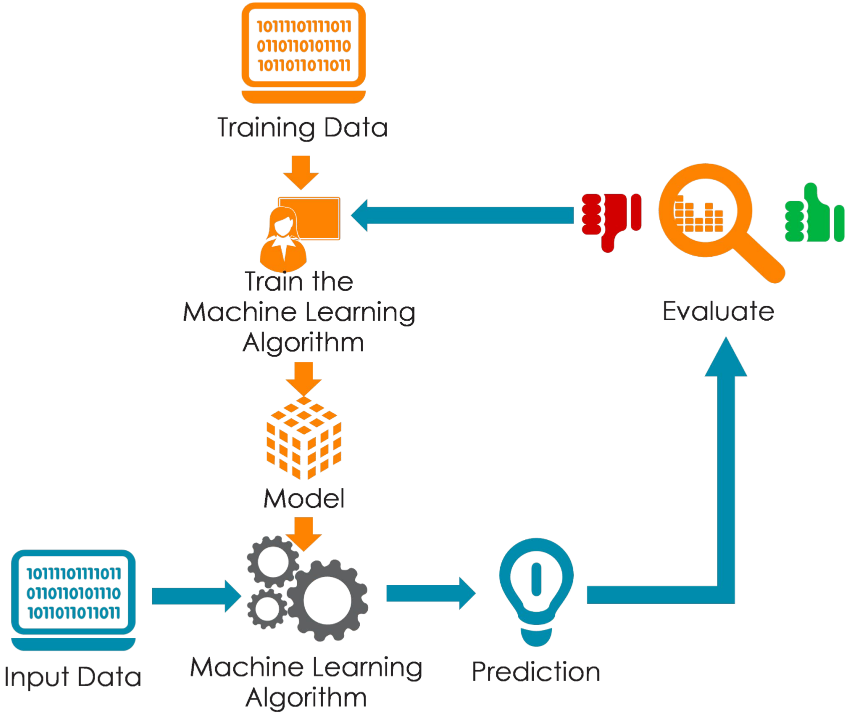
- Language is complex, varied and ambiguous.
    - There could be a gap between the literal sense and the contextual sense implied by a set of word in a particular language.
- NLP is an interdisciplinary field: Computer Science and Linguistics.
- NLP as a field is most focused upon investigating into two concomitant lines of work which both coexist as well as enhance each other.
    - NLU: core tasks that captures the grammar and syntax of language specific unstructured data.
    - NLG: Generate relevant content while adhering to the structural integrity and material correctness of a language.
- **Example**: Conversational AI like Siri/Alexa
    - Need to be able to process human languages in which instructions are provided to these devices
    - Correctly generate responses in human understandable languages or carrying out tasks that were originally intended by the user

Many Tasks of NLP

- Natural Language Generation
  - Machine Translation
  - Conversational AI
  - Abstractive Text Summarization
  - Question Answering

- Natural Language Understanding
  - Part-of-Speech Tagging
  - Coreference resolution
  - Information Extraction with Named Entities
    - Named Entity Recognition and Disambiguation
  - Text Classification
    - Sentiment Analysis
    - Spam Filtering
  - Extractive Text Summarization



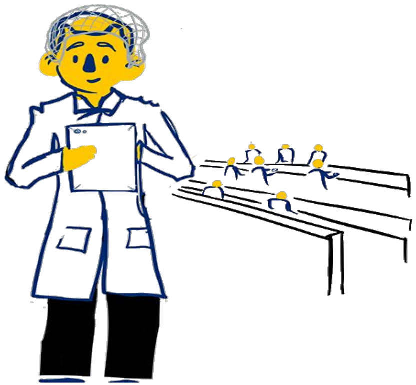


What does a NLP pipeline look like?

- Massive amount of unstructured text data available for training and evaluation
- Text Preprocessing and Feature Engineering
- Humans label this data for specific downstream task.
- Supervised learning : model trained using labelled data
    - Machine learning
    - Deep Learning
    - Heuristics
- Model Finetuning and Evaluation
- Deployment

Prerequisites

- Linear Algebra
- Statistics
- Software Engineering: Python
  - Pandas
  - Numpy
- Machine Learning
  - Scikit Learn
- Deep Learning
  - Pytorch
  - TensorFlow
- Information Retrieval

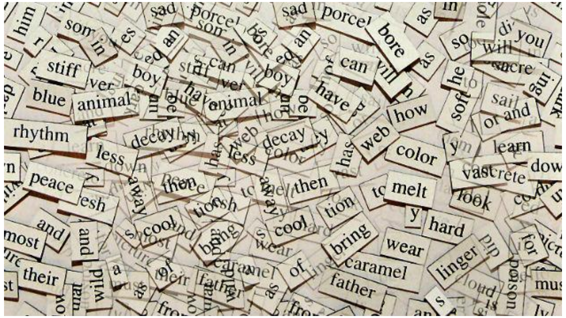


Building Blocks of Text

- **Corpora**: NLP applications rely on digitized (computer readable) collections of text or speech for learning
  - The popular Brown corpus[1] is a collection of samples from 500 written English texts from different genres (newspaper, fiction, non-fiction, academic, etc.)
  - Assembled at Brown University in 1963–64
- Fundamental unit of text processing:
  - **Tokens/Words** : Popular Brown corpus is a 1M-word collection
  - Sometimes **sentences** depending on application

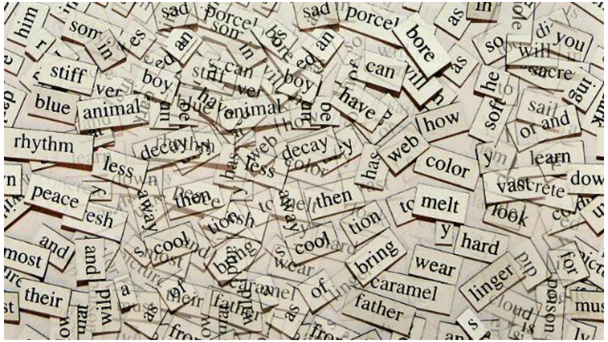
He stepped out into the hall, was delighted to encounter a water brother. They picnicked by the pool, then lay back on the grass and looked at the stars.

How many words in this excerpt?



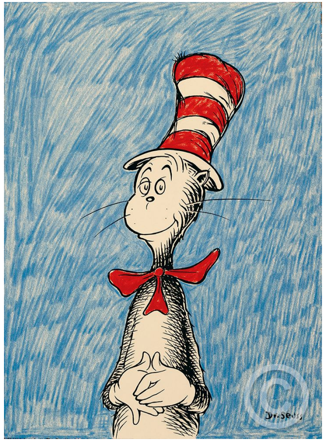
Building Blocks of Text

- Units of text processing:
  - **Word Types** – Number of unique words in the corpus
    - Set of word types is called a **Vocabulary**
    - Number of types is the Vocabulary size $|V|$
  - **Word instances** or **Tokens** are the total number N of running words.
  - **Heap's Law** – $|V| = kN^{\beta}; 0 < \beta < 1$
    - β depends on corpus size and genre; $\sim .67\ to\ .75$
- We still have decisions to make!
  - Should text be treated capitalized or uncapitalized?
    - *They* and *they* might be the same word type for the latter
  - Do we care about the ordering of words?
  - A word can be both a noun and a verb depending on context



Building Blocks of Text

- Units of text processing:
    - **Lemma** – set of lexical forms having the same stem, the same part-of-speech, and the same word sense
    - **Wordform** is the full inflected or derived form of the word.
- Wordforms are sufficient for Text Processing in English but morphologically complex languages like Arabic require Lemmatization!
- For most text processing applications, we do not use words as the unit of computation.
    - Tokenize input strings into tokens!
        - Could be words or only parts of words i.e., subwords.

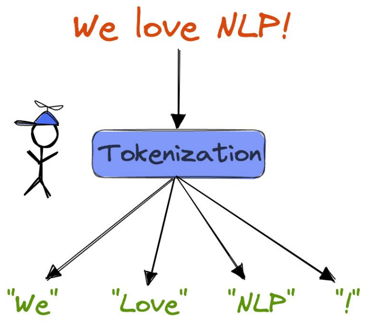


Seuss's cat in the hat is different from other cats!

Same lemma, different wordforms

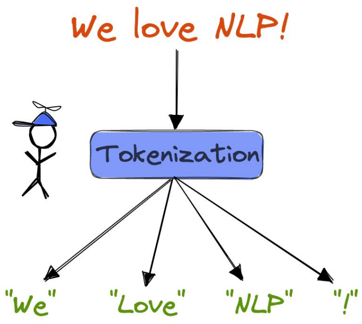
Text Preprocessing and Normalization

- Every NLP task requires preprocessing and normalization
  - Tokenizing words
  - Normalizing word formats
  - Segmenting Sentences
- **Tokenization**
  - Space-based tokenization - Segment off a token between instances of spaces
    - Simple and effective for languages that use space between words: Arabic, Cyrillic, Greek, Latin, etc.
  - Issues in Tokenization
    - Cannot blindly remove punctuations so how to deal with sentences?
      - Ph.D., AT&T, $45.55, 01/02/1996
    - Should multiword expressions be words?
      - New York, rock ‘n’ roll
    - Ciltic: a word that doesn’t stand on its own
      - are in we’re



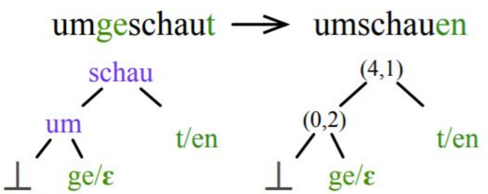
Tokenization in languages

- Many languages like Chinese or Japanese do not use spaces to separate words: Linguistic knowledge matters!
  - Chinese:
    - Chinese words are composed of characters called "hanzi" (or sometimes just "zi")
    - Each one represents a meaning unit called a morpheme.
      Each word has on average 2.4 of them.
    - 姚明进入总决赛 : 姚明 || 进入 || 总决赛 : “Yao Ming reaches the finals”
  - Other languages require more complex segmentation
    - Neural Sequence models trained by supervised machine learning required.
    - The data tells us how to tokenize.
      - Subword tokenization : Tokens can be parts of words
        as well as whole words
        - Byte-Pair Encoding
        - Unigram Language Modeling



Word Normalization Issues

- Words/tokens need to be in a standard format
    - Examples: U.S.A. or USA, or am/is/be/are
- Case Folding: Reduce all letters to lower case.
    - Some Exceptions: General Motors, US vs us, SAIL vs sail
    - Capitalization useful in some language-specific applications: Named Entity Recognition in English
- **Lemmatizaton**: Represent all words as their shared root or lemma
    - Examples:
        - am/is/be/are → be
        - He is reading detective stories → *He be read detective story*
    - Lemmatization is done by Morphological Parsing
        - Morphemes are small meaningful units that make up words
            - Stems : meaning-bearing unit; Affixes : prefix/suffix often with grammatical functions
            - Example: Parse *cats* into two morphemes *cat* and *s*



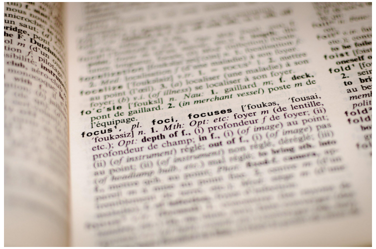
Word Normalization Issues

- Dealing with morphology can be complex in many languages.
- **Stemming**: Reduce terms to stems, chopping off affixes crudely
  - Porter Stemmer: Rule based stemming
    - ATIONAL →ATE (e.g., relational ! relate)
    - ING → if stem contains vowel (e.g., motoring ! motor)
    - SSES ! SS (e.g., grasses ! grass)

Original text:
Document will describe marketing strategies carried out by U.S. companies for their agricultural chemicals, report predictions for market share of such chemicals, or report market statistics for agrochemicals, pesticide, herbicide, fungicide, insecticide, fertilizer, predicted sales, market share, stimulate demand, price cut, volume of sales.

Porter stemmer:
document describ market strategi carri compani agricultur chemic report predict market share chemic report market statist agrochem pesticid herbicid fungicid insecticid fertil predict sale market share stimul demand price cut volum sale

Produces unrecognizable words as tokens.



Sentence Segmentation

- Contextual Unit of Text : context is important to retain, and words don't have it.
- !, ? mostly unambiguous but period "." is very ambiguous
  - Sentence boundary
  - Abbreviations like Inc. or Dr.
  - Numbers like .02% or 4.3
- Common algorithm: Tokenize first: use rules or ML to classify a period as either (a) part of the word or (b) a sentence-boundary.
  - An abbreviation dictionary can help.
- Sentence segmentation can then often be done by rules based on this tokenization.